



Introduction to

Power BI

Overview & Data Connection

Chapter 1

Introduction to Power BI

Desktop · Service · Mobile

Welcome to the world of Business Intelligence! In this chapter, you will be introduced to Microsoft Power BI — one of the most widely used data analysis and visualization tools in the industry. Whether you are a business analyst, a student, or a professional who works with data, Power BI helps you transform raw numbers into meaningful visual stories. By the end of this chapter, you will understand what Power BI is, how its three components work together, and why it has become the preferred tool for data-driven decision-making worldwide.

1.1 What is Power BI?

Power BI

A business analytics service developed by Microsoft that enables users to connect to data sources, transform and clean data, build visual reports, and share insights across an organization — all without writing complex code.

Think of Power BI as a smart reporting system for your organization. Instead of manually building charts in a spreadsheet every week, Power BI can automatically pull fresh data, refresh your visuals, and distribute the latest reports to every stakeholder — all within minutes.



Analogy — Understanding Power BI

Imagine you manage a chain of five coffee shops. Every morning, your store managers send you sales data in separate Excel files. You then manually combine them, create charts in a spreadsheet, and share a summary email. This process takes 2 hours daily.

With Power BI, you connect directly to each store's data source. Power BI automatically combines the data, generates visual charts, and publishes a live dashboard that updates every morning — saving you 2 hours every single day.

This is the core value of Power BI: automating and visualizing data so that you can focus on making decisions, not preparing reports.

1.2 The Three Components of Power BI

Power BI is not just a single application. It is a complete ecosystem consisting of three interconnected tools, each designed for a specific purpose. Understanding when to use each one is essential to working effectively with Power BI.

Power BI Desktop

Power BI Desktop

A free Windows application that is installed on your computer. It is the primary tool used to connect to data sources, transform data, build data models, and design reports. All core report development happens here.

Power BI Desktop is where all the design and development work takes place. It is the equivalent of a workshop — you bring your raw data in, clean it, arrange it, and build the final report. Once the report is ready, you can publish it to the Power BI Service for others to view.



Note

Power BI Desktop is completely free to download from the Microsoft website (powerbi.microsoft.com) or from the Microsoft Store. No subscription is required to build and save reports on your local machine.

Power BI Service

Power BI Service

A cloud-based platform (accessed via a web browser at app.powerbi.com) where reports built in Power BI Desktop are published, shared, and collaboratively managed. It requires a Power BI account (free or Pro).

Once a report is built in Desktop, it is published to the Power BI Service. Think of the Service as the publishing and distribution center. Stakeholders across an organization can log in through a browser and view the latest reports — without needing Power BI Desktop installed.

Power BI Mobile

Power BI Mobile	A free mobile application available for iOS (iPhone/iPad) and Android devices. It allows users to view and interact with reports and dashboards published to the Power BI Service while on the move.
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Power BI Mobile is designed for consumption — viewing reports and dashboards — rather than building them. A regional sales manager, for example, can open the Power BI app on their phone each morning and review yesterday's sales performance before walking into a team meeting.

Component Summary

Component	Platform	Primary Purpose	Cost
Power BI Desktop	Windows (Installed App)	Build reports and data models	Free
Power BI Service	Web Browser (Cloud)	Publish, share, and collaborate	Free / Pro / Premium
Power BI Mobile	iOS / Android (App)	View reports on the go	Free

 **Pro Tip**

As a beginner, you will spend most of your time in Power BI Desktop. Once you are comfortable building reports, you will then explore the Power BI Service to share them. Power BI Mobile is something you can explore later as an additional skill.

1.3 Why Power BI? — Industry Adoption

Before learning how to use a tool, it is valuable to understand why it is worth learning. Power BI has become one of the most in-demand skills in the data and analytics industry. Here are the key reasons for its widespread adoption.

Reason	Explanation
No programming required	Anyone who can use Excel can learn to build reports in Power BI. No SQL or coding knowledge is mandatory for most tasks.
Works with almost any data source	Power BI connects to over 100 data sources — Excel, SQL Server, Google Sheets, Salesforce, SAP, and many more.

Reason	Explanation
Live and automatic data refresh	Reports can be set to automatically refresh data on a schedule, ensuring stakeholders always see the latest information.
Industry-standard tool	Power BI consistently appears in top data analyst job descriptions. It is used by over 250,000 organizations worldwide.
Tight integration with Microsoft Office	For organizations already using Excel and Teams, Power BI integrates naturally into existing workflows.
Cost-effective	The Desktop application is free. Pro licenses (for sharing) are significantly more affordable than competing tools.



Section 2

The Power BI Desktop Interface

Understanding the workspace before you build

2.1 Overview of the Power BI Desktop Interface

Before you start connecting to data or building reports, it is important to familiarize yourself with the layout of the Power BI Desktop interface. Understanding where each feature is located will make you significantly more efficient when you start working on real projects.

The Ribbon (Menu Bar)

Located at the very top of the screen, the ribbon is organized into tabs — Home, Insert, Modeling, View, Optimize, and Help. Each tab contains grouped buttons for related actions. Most frequently used actions, such as importing data, transforming data, and publishing, are found under the Home tab.

The Three Views — Report, Data, and Model

On the left side of the screen, you will find three icons that switch between the three core views of Power BI Desktop. Each view serves a distinct purpose in the report-building process.

View	Icon Description	Purpose
Report View	Bar chart icon	This is where you design reports by adding visuals (charts, tables, maps) to a canvas. This is the view you will use most often.
Data View	Grid/table icon	Displays your data in a spreadsheet-like grid. Use this to inspect the actual values in your data, check for errors, or add calculated columns.
Model View	Diagram/network icon	Shows the relationships between multiple data tables. Used when working with more than one table in a data model.

The Report Canvas

The large blank white area in the center of the screen is the Report Canvas. This is where your visual reports are built. You drag and drop visual elements — charts, tables, cards, maps — onto this canvas to design a report page. Each report can have multiple pages, similar to multiple sheets in an Excel workbook.

The Visualizations Pane

Located on the right side of the screen, the Visualizations Pane is where you choose the type of visual you want to create — such as a bar chart, pie chart, line chart, map, or table. Once a visual type is selected, the pane also shows three sub-sections: Fields (what data to show), Format (visual styling), and Analytics (trend lines and reference lines).

The Fields Pane

Also on the right side (to the right of the Visualizations Pane), the Fields Pane lists all the tables and columns available in your data model. When you load data into Power BI, each table appears here with an expandable list of its columns. You drag fields from this pane onto the canvas or into the Visualizations Pane to build your reports.



Pro Tip

If you cannot find a pane on your screen, it may have been accidentally closed. Go to the View tab in the Ribbon and check that 'Visualizations', 'Fields', and 'Filters' are enabled. You can also drag panes to resize them.

Section 3

Data Sources in Power BI

Connecting Power BI to Your Data

3.1 What is a Data Source?

Data Source

Any system, file, or platform that contains data which Power BI can connect to and retrieve for analysis. A data source can be as simple as an Excel file on your desktop or as complex as a live corporate database server.

In real-world organizations, data rarely lives in just one place. Sales figures may be in a SQL database. Employee records may be in an HR system. Marketing data may be in a cloud platform. Power BI's ability to connect to all these sources simultaneously — and combine them into a single report — is one of its most powerful capabilities.

**Analogy — Data Sources**

Think of Power BI as a report-writing assistant. Before your assistant can write a report, you need to tell them where to find the information. You might say: 'The sales numbers are in this Excel file. The customer list is in our company database. The website traffic stats are on our web analytics platform.'

Each of those locations — the Excel file, the database, the website — is a data source. Power BI is the assistant that goes to each location, gathers the data, and brings it all together in one place.

3.2 Commonly Used Data Sources

Power BI supports over 100 data sources. As a beginner, you will most frequently work with four of the most common ones. Each is described below with a practical professional example.

Excel Workbooks (.xlsx / .xls)

Excel Data Source

A Microsoft Excel file (.xlsx or .xls) that contains data in one or more worksheets or tables. Power BI can read data directly from any structured worksheet or Excel Table within the file.

Excel is the most common starting point for Power BI users. Most organizations have years of historical data stored in Excel files. Power BI can connect to these files and automatically bring in the data, saving you from copying and pasting manually.

Professional Example — Excel Source

A Finance team maintains a monthly budget tracker in an Excel file called 'Budget_FY2025.xlsx'. The file has three sheets: Revenue, Expenses, and Summary.

By connecting this file to Power BI, the analyst can build a live dashboard that shows budget vs actual performance. Whenever the Excel file is updated, the Power BI report can be refreshed to reflect the new numbers — instantly.

CSV Files (.csv — Comma Separated Values)

CSV File

A plain text file where each row represents one record and each value within that row is separated by a comma. CSV files are universally supported and commonly used to export data from databases, ERP systems, and web platforms.

CSV files are the universal data exchange format. When you export a report from any software — whether it is Tally, SAP, Salesforce, or an e-commerce platform — it almost always gives you the option to export as a CSV file. Power BI can connect to and read these files with ease.

Professional Example — CSV Source

An e-commerce company exports a daily order log from their platform as a CSV file named 'Orders_2025-01-15.csv'. Each row in the file represents one order, with columns for Order ID, Customer Name, Product, Quantity, and Total Amount.

By connecting this CSV file to Power BI, the analyst can build a daily sales report without manually reformatting the exported data.

Pro Tip

CSV files do not support formatting, formulas, or multiple sheets — they contain only raw data. This actually makes them very reliable for Power BI, since there is no risk of accidentally importing a formula instead of a value.

SQL Server Database

SQL Database

A structured database management system (DBMS) such as Microsoft SQL Server, MySQL, or PostgreSQL, where data is stored in organized tables with defined relationships. Power BI can connect directly to the database server to retrieve data.

Most medium and large organizations store their transactional data in a SQL database rather than individual files. This includes records such as customer orders, inventory movements, employee records, and financial transactions. Connecting Power BI directly to a SQL database means your reports are always reading from the single source of truth.

Professional Example — SQL Source

A manufacturing company has a SQL Server database named 'ProductionDB'. It contains tables for raw material inventory, production orders, machine utilization, and quality control results.

The Production Manager connects Power BI to this database. Every morning, the report automatically shows current inventory levels, production efficiency for the previous day, and any quality issues flagged overnight — all pulled live from the database.

Note

To connect to a SQL database, you will need the server name (or IP address) and database name from your IT team or database administrator. You may also need a username and password if the database requires authentication.

CLOUD & AI ACADEMY**Web and SharePoint Sources****Web Data Source**

A URL-accessible data source on the internet or company intranet — such as a webpage with a data table, a SharePoint list, or an API endpoint — that Power BI can connect to and retrieve data from in real time.

Power BI can also pull data directly from websites and cloud platforms. This is useful when working with publicly available data — such as government statistics, exchange rates, or market data — or when your organization stores data in SharePoint or Microsoft 365.

Professional Example — Web Source

An Import-Export company needs to track live currency exchange rates. Rather than manually checking a website each day, the analyst connects Power BI to a financial data URL that publishes daily exchange rates.

Power BI pulls the latest rates automatically, and the report recalculates all international transaction values in the local currency without any manual intervention.

Data Sources — Quick Reference

Data Source	File/Platform	Best Suited For	Skill Level
Excel Workbook	.xlsx / .xls	Departmental reports, budget trackers, structured lists	Beginner
CSV File	.csv	Exported data from apps, ERP systems, e-commerce platforms	Beginner
SQL Server	Database server	Enterprise transactional data, large structured datasets	Intermediate
Web / SharePoint	URL / Cloud	Live public data, shared organizational files	Intermediate
Other Sources	100+ options	SAP, Salesforce, Google Analytics, Azure, and many more	Advanced

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Section 4

Loading and Previewing Data

Connecting to Your First Data Source

4.1 Using 'Get Data' to Connect a Source

The first step in every Power BI project is connecting to your data. Power BI calls this process 'Get Data'. This section walks you through the process step by step, using an Excel file as the example — since this is the most common starting point for beginners.

Step-by-Step: Connecting to an Excel File

Step 1 — Open Power BI Desktop

Launch Power BI Desktop from the Start menu or the Microsoft Store. When it opens, you will see a splash screen with options to open a recent file, view samples, or get started. Close or dismiss this screen to see the blank canvas.

Step 2 — Click 'Get Data' on the Home Ribbon

In the Home tab of the ribbon at the top of the screen, locate and click the 'Get Data' button. A dropdown may appear with common sources, or you can click 'More...' to see the full list of all available data sources.

Step 3 — Select Excel Workbook

From the Get Data dialog or dropdown, select 'Excel Workbook'. A file browser window will open, allowing you to navigate to the location of your Excel file on your computer. Select the file and click 'Open'.

Step 4 — Navigate the Navigator Window

After selecting your file, the Navigator window appears. This shows all the worksheets and tables found in the Excel file. On the left side, you will see a list of all available sheets. When you click on a sheet name, a preview of its data appears on the right side of the window.



What is the Navigator Window?

The Navigator window acts like a table of contents for your data file. Before committing to loading all the data, it lets you preview each sheet or table to make sure you are selecting the correct one.

For example, an Excel file might have 5 sheets: Sales Data, Reference List, Instructions, Archive 2022, and Summary. You would only want to load 'Sales Data' — the Navigator lets you confirm this before proceeding.

Step 5 — Load or Transform

At the bottom of the Navigator window, you will see two buttons: 'Load' and 'Transform Data'.

Load	Transform Data
Loads the data directly into Power BI as-is.	Opens Power Query Editor before loading — allowing you to clean and reshape the data first.
Use this when your data is already clean and well-structured.	Use this when the data needs modifications, such as removing blank rows, renaming columns, or changing data types.
Faster and simpler for beginners.	Recommended for professional use and real datasets.

For your first exercise, click 'Load' to import the data directly. The data will load into the Power BI data model and you will return to the Report Canvas. The Fields Pane on the right will now show the table and its columns, confirming that the data has been successfully imported.

4.2 Previewing Loaded Data in Data View

After loading data, it is good practice to review it in the Data View before building any visuals. This helps you verify that the data has been imported correctly — that all rows and columns are present, values look correct, and data types are appropriate.

How to Access Data View

On the left side of the Power BI Desktop window, click the Data View icon (the grid/table icon). The screen will switch from the canvas to a spreadsheet-like view of your data. You can scroll horizontally and vertically to inspect all columns and rows.

What to Check	Why It Matters
All columns are present	Confirms Power BI imported all the fields from your source correctly.
Column names are correct	Ensures column headers were read properly and not merged or split incorrectly.
Data types are correctly assigned	Dates should appear as dates, numbers as numbers, and text as text.

What to Check	Why It Matters
No unexpected blank rows	Extra blank rows can cause errors in calculations and filtering.
Values look reasonable	A quick sanity check — sales figures of 1,000,000 in a small business report may indicate a formatting issue.



Section 5

Import vs DirectQuery

Choosing How Power BI Reads Your Data

5.1 Two Ways to Connect — Import and DirectQuery

When you connect Power BI to a data source — particularly a database — you are asked to choose how the data should be accessed. There are two primary connection modes: Import and DirectQuery. Choosing the right mode depends on the nature of your data and how frequently it changes.

Import Mode

Data is copied from the source and stored inside the Power BI file (.pbix) on your local machine or in the Power BI Service. All calculations and filtering happen on this local copy.

DirectQuery Mode

No data is copied or stored. Every time a visual is viewed or refreshed, Power BI sends a live query to the original data source and retrieves the latest data in real time.

**Analogy — Import vs DirectQuery**

Imagine you need to reference a textbook for a project.

Import Mode is like photocopying the relevant chapters and working from your own copy. The copy is instantly accessible, but it reflects the state of the book at the time you photocopied it. If the textbook is updated later, your copy remains unchanged.

DirectQuery Mode is like going back to the library every time you need information. The information is always current and up-to-date, but each visit takes a little more time than reading from your own copy.

5.2 When to Use Each Mode

Import Mode	DirectQuery Mode
Data is loaded and stored in the .pbix file.	Data stays in the source system — nothing is stored in Power BI.

Best for datasets up to several hundred MB.	Best for very large datasets that cannot be stored locally.
Report loads and filters very quickly.	Each interaction queries the source — can be slower.
Data is a snapshot; must be manually or scheduled refreshed.	Always shows real-time or near-real-time data.
Works with all data sources including Excel and CSV.	Only supported for databases and select sources (not Excel or CSV).
Best for: Historical analysis, batch-updated data.	Best for: Live dashboards, operational reporting.

 **Note**

As a beginner, you will primarily use Import Mode. When you load data from Excel or CSV files, Import is the only available option — DirectQuery is not applicable to file-based sources. DirectQuery becomes relevant when you start connecting to live databases.

Practical Example

Consider two different reporting scenarios to understand when each mode is most appropriate:

Scenario	Recommended Mode	Reason
Monthly Sales Summary Report — data updated once a month from an Excel export	Import Mode	Data changes infrequently. Import mode provides fast performance and offline access.
Live Operations Dashboard — hospital patient admissions updated every 5 minutes in a SQL database	DirectQuery Mode	Data is continuously updated. DirectQuery ensures the dashboard always reflects current status.
Annual Financial Audit Report — 3 years of historical data from a large SQL data warehouse	Import Mode	Historical data does not change. Import provides faster report interaction than live queries.

Section 6

Basic Data Model Introduction

How Tables Relate to Each Other in Power BI

6.1 What is a Data Model?

Data Model	The organized collection of tables and the defined relationships between those tables within a Power BI report. A well-structured data model enables accurate, efficient calculations and reporting across multiple related datasets.
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When you load more than one table into Power BI, they do not automatically know how they relate to each other. A data model is the framework that defines those connections. Think of it as telling Power BI: 'These two tables are linked — the Customer ID in the Sales table corresponds to the Customer ID in the Customers table.'

 Analogy — The Data Model
Consider the administration office of a university. They have three separate registers:
1. A Students Register — contains Student ID, Name, Department, Year of Enrollment.
2. An Attendance Register — contains Student ID, Date, Present/Absent.
3. A Marks Register — contains Student ID, Subject, Marks Obtained.
Each register is a separate table. But notice that they are all connected through the 'Student ID'. If you want to see attendance and marks for a specific student by name, you need to link all three registers using Student ID as the common key.
In Power BI, this linking process is called creating a Relationship, and the combined structure of all three linked tables is called the Data Model.

6.2 Understanding Relationships

A relationship in Power BI is a link between two tables, defined by a common column that appears in both tables. This common column is called the key. Relationships allow Power BI to correctly filter and aggregate data across multiple tables when building visuals and writing calculations.

Types of Relationships

Relationship Type	Description	Example
One-to-Many (1:*)	The most common type. One row in Table A relates to multiple rows in Table B.	One Customer can have many Orders. One Product can appear in many Sales records.
Many-to-One (*:1)	Identical to One-to-Many, just viewed from the other direction.	Many Sales records relate to one Product.
One-to-One (1:1)	One row in Table A corresponds to exactly one row in Table B.	One Employee record corresponds to one Payroll record.

Viewing Relationships in Model View

To view and manage relationships, switch to the Model View by clicking the diagram icon on the left panel. Each loaded table appears as a box showing its column names. Lines connecting the tables represent relationships, with small symbols indicating the relationship type (1 and * for One-to-Many).

 **Pro Tip**

When you load multiple tables from the same Excel file or database, Power BI often automatically detects relationships based on column names. Always verify these auto-detected relationships in Model View before building your report, as they may occasionally need adjustment.

Section 7

Practice Exercise

Applying What You Have Learned

7.1 Exercise Overview

This exercise walks you through the complete workflow covered in this chapter — from opening Power BI Desktop to loading data and exploring the interface. Follow each step carefully. The goal is not to build a complete report yet, but to become comfortable with the process of connecting to data and navigating the Power BI interface.



Before You Begin

Ensure Power BI Desktop is installed on your computer. If not, download it for free from:
<https://powerbi.microsoft.com/desktop>

Your instructor will provide the practice Excel file: 'SalesData_Practice.xlsx'. Save this file to your Desktop or a folder you can easily navigate to.

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7.2 Step-by-Step Exercise Instructions

Exercise 1 — Exploring the Interface

Open Power BI Desktop. Without loading any data, take 5 minutes to explore each part of the interface:

- Identify the **Ribbon** and locate the Home, Insert, and Modeling tabs.
- Find the three **View icons** on the left panel (Report, Data, Model).
- Locate the **Visualizations Pane** and the **Fields Pane** on the right side.
- Identify the **Report Canvas** and the **Pages strip** at the bottom.
- Click on each view icon (Report, Data, Model) and note how the center area changes.

Exercise 2 — Loading Excel Data

Follow these steps to load the practice Excel file into Power BI:

- Click **Get Data** on the Home Ribbon.
- Select **Excel Workbook** from the list.

- Navigate to the 'SalesData_Practice.xlsx' file and click Open.
- In the Navigator window, click on the **Sales** sheet to preview its data.
- Verify the data preview looks correct, then click **Load**.
- Confirm the table appears in the Fields Pane on the right.

Exercise 3 — Inspecting Data in Data View

After loading the data, switch to Data View and answer the following questions:

- How many rows (records) does the Sales table contain?
- How many columns are in the table? What are their names?
- What data type has Power BI assigned to the 'Order Date' column? (Look at the icon next to the column name.)
- Are there any obviously blank or incorrect values in any column?

Exercise 4 — Exploring the Model View

Switch to Model View and observe the following:

- Is there more than one table shown? If only one table was loaded, only one box will appear.
- Hover over the table box — notice it shows the table name and column list.
- If your instructor asks you to load the second sheet 'Products' as well, do so using Get Data. Then check if Power BI automatically creates a relationship between the two tables.



Exercise Completion Checklist

Before moving to the next chapter, confirm you can do the following without assistance:

- ✓ Open Power BI Desktop and identify all interface components
- ✓ Connect to an Excel file using Get Data
- ✓ Navigate the Navigator window and load a sheet
- ✓ Switch between Report View, Data View, and Model View
- ✓ Locate loaded tables in the Fields Pane

Chapter Summary

In this chapter, you have been introduced to the foundational concepts of Power BI. Here is a recap of all the key topics covered:

- **Power BI** is a business analytics platform by Microsoft used to connect to data, build visual reports, and share insights across organizations.
- It consists of three components: **Power BI Desktop** (for building), **Power BI Service** (for publishing and sharing), and **Power BI Mobile** (for viewing on the go).
- The **Power BI Desktop interface** includes the Ribbon, the three view icons (Report, Data, Model), the Report Canvas, the Visualizations Pane, and the Fields Pane.
- A **Data Source** is any system or file from which Power BI retrieves data. Common sources include Excel workbooks, CSV files, SQL Server databases, and web/cloud platforms.
- The **Get Data** button on the Home Ribbon is used to connect Power BI to any supported data source.
- **Import Mode** copies data into the Power BI file for fast performance. **DirectQuery Mode** reads live data from the source without storing a copy.
- A **Data Model** is the organized set of tables and their defined relationships. Relationships link tables through a common column (key) to enable cross-table analysis.

